Assignment 1 – Project Initiation Document

Submitters: Eitan Beriy.

Mohamad Ali Mousa.

Yuval Frankel.

Project Selection Reason

We chose the Barber4U project because one of our team members knows a barbershop chain that still manages all appointments by phone. This method is slow, uncomfortable, and often causes mistakes in scheduling. The barbershop owner also plans to open more branches in different locations, which will make phone-based scheduling even harder.

Because of this, we decided to create a smart mobile app that helps customers book appointments easily and helps barbers and managers organize their work in a faster and more efficient way. The project is based on a real need, and that is why we chose it.

In addition, if the app works well, it can be offered to other barbershops in the future as a ready-to-use system, which may create business opportunities for the team.

Business Case & Project Charter

Project Name

Barber4U.

Problem Statement

Customers often find it difficult to locate nearby barbers or schedule appointments efficiently. Many barbershops still rely on walk-ins or phone calls, leading to long waits and missed opportunities. Barbers also lack a simple way to promote their work and receive feedback. Barber4U solves this by providing a mobile app that lets customers easily find branches, book appointments, and view barber portfolios, while giving barbers tools to manage schedules and display their work.

Market Research

Apps like Booksy, StyleSeat, and Fresha already exist but cater to all types of salons, not just barbershops or multi-branch chains. They can

also be expensive and complex for smaller operations. Barber4U offers a focused, affordable, and brand-specific platform designed for a single barbershop chain.

Purpose and Objectives
The purpose of Barber4U is to make the haircut booking experience smooth and accessible.

Main objectives

- Help customers find nearby branches and book easily.
- Allow barbers to upload haircut photos and get rated.
- Improve scheduling efficiency and reduce no-shows.
- Strengthening brand visibility and customer loyalty.

Benefits and Impact
The app will enhance customer convenience, reduce administrative effort, and help barbers build a reputation. The barbershop chain will gain stronger branding, customer retention, and useful data on performance and engagement.

Scope

The project includes developing a mobile app with features for booking, GPS branch locator, barber profiles, photo galleries, and ratings.

High-Level Risks

Main risks include technical challenges in syncing bookings, low early adoption, and data privacy concerns. These will be addressed through reliable backend architecture, promotional campaigns, and secure authentication practices.

Alternative Solutions

Alternatives like manual bookings or third-party apps lack customization and integration. Developing a dedicated mobile app offers the best balance of control, branding, and user experience.

Key Stakeholders

Project sponsor (barbershop management), project manager, development team, barbers, customers, and marketing staff.

Resources

Required resources include app developers, a backend specialist, a UI/UX designer, and testing devices. Technologies such as Firebase or AWS will support hosting, with Google Maps API for location services.

Statement of Work

Vision Statement

The vision of Barber4U is to create a simple, easy-to-use, and modern mobile app that helps customers choose a branch from a barbershop chain, see all available barbers, check free time slots, and book an appointment with one click. The app will also allow barbers to manage their schedule, approve or reject appointment requests, and suggest another time when needed. The goal is to provide a faster, more organized, and more convenient experience for both customers and barbers, instead of using phone calls and manual coordination.

Scope of Project

The project includes the development of an Android application written in Java. The app supports three types of users: customers, barbers, and the system manager (admin).

For customers, the app will allow registration and login, choosing a branch, viewing the list of barbers, checking available time slots, viewing the gallery of barber work, and booking appointments.

For barbers, the app will include a schedule management screen, viewing appointment requests, approving or rejecting appointments, and offering another time when needed. Barbers will also be able to upload pictures and manage their personal gallery.

For the admin, the project will include an advanced management system that allows managing branches, barbers, and appointments, controlling user permissions, and monitoring general activity in the system.

The app will use two main device features: **the camera**, for uploading images and creating barber galleries, and **location services**, using Google Maps to show the location of branches and help users navigate to them. The project will also use Firebase for data storage, authentication, and managing user information.

The project **does not include** a website, iOS support, a payment system, physical inventory management, or advanced business reports.

Scope of Work

The project will include several main stages.
In the first stage, the team will collect requirements and plan the system, including defining the user types, designing the app screens, and planning the Firebase database. The team will also use Git for version control and for sharing work between team members.

In the second stage, the UI/UX design of the main app screens will be created.

During the development stage, all parts of the Android app will be built in Java, including the login and registration system, branch selection, barber selection, availability display, appointment booking, the barber management interface, the admin management interface, image upload using the camera, and location services with Google Maps.

In the integration stage, the app will be fully connected to Firebase, including user management, data storage, and image handling.
After that, functional tests, permission tests, and user flow tests will be performed. Finally, bugs will be fixed, improvements will be made, and the team will prepare the final version for the presentation and submission.

Key Features

The app will include several main features.
For customers: registration and login, choosing a branch, viewing the list of barbers, viewing the gallery of barber work, checking available time slots, and booking appointments.

For barbers: managing their schedule, viewing appointment requests, approving or rejecting appointments, offering another time when needed, and uploading pictures to their work gallery using the camera.

For the admin: managing branches, managing barbers, managing users, viewing all appointments, and controlling system permissions.

In addition, the app will use the camera for uploading images, location services with Google Maps for showing branch locations, and Firebase for data storage, user authentication, and handling system information.

Constraints

The project is developed under several main constraints.
In terms of time, the project must be completed during the semester, with weekly progress presentations in class and a final submission deadline.

In terms of knowledge, the development team needs to work with Android in Java, use Firebase, and implement camera features and location services, which require learning and practice during the project.

In terms of technology, the app is designed only for Android and does not include iOS support. The system must use Firebase for data storage and Google Maps for showing branch locations.

In terms of resources, the team has only three developers, so good coordination through Git and meeting the project timeline are required.
There are also hardware limitations, since testing must be done on different Android devices with different camera quality and different levels of GPS accuracy.

Dependencies

The project depends on several external and technical factors.
On the service level, the system uses Firebase for user authentication, data storage, and image storage, and Google Maps and GPS services for showing the location of branches. The app also depends on camera and location permissions, as well as the Camera API and Location Services on Android devices.

The system requires an active internet connection for loading data and uploading images.
In terms of tools, the development uses Android Studio, Java, Gradle, and Git for version control.
The success of the project also depends on good teamwork, coordination between the three developers, and completing the assigned tasks on time.

Deliverables

The main deliverables of the project will include a full Android application in APK format, containing all the main features: user registration and login, branch selection, barber selection, appointment booking, the barber interface, the admin interface, camera support, location services, and full integration with Firebase.

In addition, the full source code of the project will be submitted through a GitHub repository, together with all project documents, including the Project Initiation Document (PID), the SOW document, and any other required documentation.
Weekly progress presentations will also be given in class during the semester.
At the end of the project, a final presentation and a live demo of the system will be delivered.

A basic testing report describing the tests performed and the fixes applied will also be submitted.

Feasibility Study Report

Timeline

The project timeline is planned for the full semester, which includes 13 weeks of work.

During weeks 1–2, the team completed the requirement gathering, the initial system planning, and the planning of the Firebase database.

In weeks 3–4, the team will focus on designing the app screens (UI/UX), completing the project documents, and presenting the weekly progress in class.

In weeks 5–7, the main user screens will be developed, including login, registration, branch selection, barber list, gallery view, and checking available time slots.

In weeks 8–9, the management interfaces for barbers and the admin will be developed, including image uploads and basic Firebase connections.

In weeks 10–11, full integration with Firebase will be completed, and Google Maps and location services will be added.

In week 12, testing will be performed, bugs will be fixed, and the user experience will be improved.

In week 13, the team will prepare the final demo, build the final presentation, and submit the project.

Throughout the whole semester, weekly progress presentations will be given in class as required by the course.

Performance Criteria

The success of the project will be measured by several main criteria.
For functionality, the app will be considered successful if users can register and log in, choose a branch and a barber, check available time slots, and book an appointment. Barbers should be able to approve or reject appointments and upload photos to their personal gallery. The admin should be able to manage branches, barbers, and appointments through the admin interface.

The app must successfully integrate location services with Google Maps and use the camera for uploading images.
In terms of software quality, the app should be stable, without crashes or critical errors, and should provide a clear and easy user experience.
The project will also be considered successful if it meets the timeline and includes a full working demo at the end of the semester.

Risk Management

The project includes several main risks and ways to handle them.

Risk 1: Firebase integration problems
Mitigation: testing early and working with Firebase from the beginning.
Contingency: using a simpler data structure or switching to Realtime Database if needed.

Risk 2: Camera or permission issues
Mitigation: testing on different Android devices and using proper error handling.
Contingency: allowing image upload from the gallery if the camera does not work.

Risk 3: GPS or Google Maps errors
Mitigation: testing location services and using a valid API key.
Contingency: showing location by address if GPS is not accurate.

Risk 4: Work overload or team coordination problems
Mitigation: clear task division and weekly follow-up.
Contingency: moving tasks between team members or simplifying features.

Risk 5: Critical bugs close to the submission date
Mitigation: regular testing during development.
Contingency: keeping a stable demo version ready in advance.

Stakeholder Analysis Document

Technical Feasibility

The Barber4U application is technically feasible with the resources and technologies already defined in the project's initiation document. The system requires:

● Mobile application development (Android).

● Backend development using cloud services such as Firebase for authentication, database

storage, scheduling, and image hosting.

● Google Maps API for branch location services.

● A scalable database structure to support 16 barbers across 4 branches.

● UI/UX design to ensure a smooth and intuitive user experience.

All required technologies are widely available, reliable, and well-documented. The project does not

require specialized hardware. Therefore, Barber4U is technically achievable within the given constraints.

Operational Feasibility

Barber4U provides a complete digital solution for a small barbershop chain with 4 locations. The current workflow: phone calls, walk ins, and manual scheduling, create inefficiencies such as:

● Long wait times.

● Unpredictable daily schedules for barbers.

● No structured way for customers to compare barbers or branches.

● No centralized system for managing four barbershop locations.

● Lack of customer ratings or portfolios to help customers choose barbers.

Barber4U addresses these challenges by offering:

● A unified booking system for all 16 barbers.

● Branch locator and availability information.

● Barber portfolios with customer ratings.

● Automatic schedule synchronization and reduced administrative overhead.

● Improved chain-level transparency and branding.

Because the system integrates smoothly into the daily operations of the barbershop chain and solves real

operational problems, Barber4U is operationally feasible and provides clear value.

Financial Feasibility

This section analyzes development costs, expected revenues, and long-term financial performance.

(100NIS/hour)

Development Cost (NIS)

Task Hours Cost (₪100/hr)

Requirements & Design 120 ₪12,000

UI/UX 80 ₪8,000

Mobile Development 250 ₪25,000

Backend Development 200 ₪20,000

Integration (Maps, images, auth) 70 ₪7,000

Testing 100 ₪10,000

Project Management 60 ₪6,000

Total 880 hrs ₪88,000

Total Project Cost: ₪88,000

This is within the investor's available budget of ₪100,000.

Revenue Model

Since the system serves 16 barbers, the pricing model is based on a subscription per barber.

Assumptions:

● Subscription fee: ₪150 per month per barber

● 16 barbers total

● Additional income (ads, promotions): ₪4,000 per year

Calculated Annual Revenue:

Component Amount

Subscription Revenue: 16 × ₪150 × 12 ₪28,800

Additional Income ₪4,000

Total Annual Revenue ₪32,800

Net Present Value (NPV) – 5 Years

Assumptions:

● Annual cash flow: ₪32,800

● Discount rate: 10%

● Project lifetime: 5 years

● Initial investment: ₪88,000

Present Value Calculations:

Year Cash Flow (NIS) PV Formula Present Value (NIS)

1 32,800 32,800 / 1.1 29,818

2 32,800 32,800 / 1.1^2 27,107

3 32,800 32,800 / 1.1^3 24,642

4 32,800 32,800 / 1.1^4 22,402

5 32,800 32,800 / 1.1^5 20,366

Total PV of 5 Years:

29,818 + 27,107 + 24,642 + 22,402 + 20,366

= ₪124,335

NPV Calculation:

NPV = Total PV – Initial Cost

NPV = 124,335 – 88,000 = ₪36,335

NPV is positive: +₪36,335

This means the project creates financial value for the investor over 5 years.

3.3.4 ROI Calculation

ROI = (Total Gain – Cost) / Cost × 100%

Year 1 ROI:

(32,800 – 88,000) / 88,000 × 100

= –62.7%

(negative ROI in Year 1 is normal for software projects)

5-Year ROI (without discounting):

Total revenue (5 years): 32,800 × 5 = ₪164,000

ROI = (164,000 – 88,000) / 88,000 × 100

= +86.36%

✔ Long-term ROI is strongly positive.

3.4 Financial Feasibility Conclusion

● The project fits within the investor's budget

● The revenue from 16 barbers across 4 branches generates meaningful annual income.

● The 5-year NPV is positive, indicating profitable long-term value.

● The system becomes financially beneficial even with a small chain size.

● Barber4U is financially feasible, sustainable, and strategically strong.

Stakeholder List

- Project Sponsor: Barbershop company management
- Project Manager
- Development Team: Mobile developers, backend developer, UI/UX designer
- Barbers: Staff working in the company's branches
- Customers: Users booking appointments through the app
- Marketing Team: Responsible for promotions and user engagement

Interests and Expectations

- Project Sponsor: Expects a reliable, user-friendly app that improves customer satisfaction and operational efficiency.
- Project Manager: Needs clear requirements, stable timelines, and smooth coordination between teams.
- Development Team: Expects well-defined features, access to required resources, and timely feedback.
- Barbers: Want an easy-to-use system for managing appointments, showcasing their work, and gaining ratings.
- Customers: Expect quick access to nearby branches, simple appointment booking, and the ability to view barber profiles.

- Marketing Team: Needs tools to promote the app, gather user insights, and increase overall engagement.

Communication Plan

- Weekly Team Meetings: Project manager and development team meet to review progress and address issues.
- Biweekly Sponsor Updates: Short reports or presentations to keep the sponsor informed on milestones and risks.
- Monthly Stakeholder Review: Broader meeting with marketing, barbershop management, and project manager to review features, feedback, and upcoming goals.
- In-App Feedback Channels: Allow customers and barbers to submit feedback or report issues directly through the app.
- Digital Communication Tools: Slack or email for day-to-day team communication, and shared documents via Google Drive for version control and planning.